



THOMAS LABETOUILLE

Software Developer — Applied AI & UE5 Gameplay (C++)

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Full driving licence — own vehicle

PROFILE

Systems built around locally-run language models, agent-driven editor tooling, and Unreal Engine 5 gameplay programming (C++). One thread runs through all of it: code that compiles and appears to work is not correct code — so every project ships with the means to verify it along the real execution path (regression suites, property-based tests, mutation testing, playtest agents, CI).

PROFESSIONAL EXPERIENCE

Embedded Systems Technician — CS Group

Sept. 2024 – Nov. 2025

- Designed and integrated interactive scenarios for aeronautical simulators (Inscape VTS, for national defence), working against strict functional requirements.
- Built modular software architectures, debugged the behaviour of complex simulated systems, and worked from classified documentation.
- Worked alongside domain experts (systems engineers, pilots) in a regulated defence environment.

Gameplay Programmer (Freelance) — Rainbow Ant Studio

Dec. 2023 – Sept. 2024

- Built the character's movement and interaction mechanics, a swimming system, dynamic weather integration and an interactive in-game map in Unreal Engine 5.
- Integrated animations through State Machines and Blend Spaces; wrote progress reports for a distributed team.

PERSONAL PROJECTS Nov. 2025 to present

AI Agent × UE5 — verified level-generation toolchain — [open source](#)

2025–2026

- An MCP bridge and C++/Slate plugin driving the UE5 editor from a conversation; 117 tests replayed on every change.

GameAnimationSample — UE5 tooling suite — [C++](#), [Slate](#), [Python](#)

2026

- A Slate blockout tool, Behavior Trees built by script, C++ combat AI, and a harness grading the agent itself.

Room Builder — Unity editor tools and a geometry harness — [Unity 6](#), [C#](#), [.NET](#)

2026

- 14 properties checked outside the engine across 40,000 panels; mutation testing: 21 defects injected, 19 caught.

Comptoir — holiday search assistant — [Python](#), [local LLM](#), [SQL](#), [Java](#)

2026

- The model understands and writes but decides nothing: the verifiable is settled in Python. 167 tests, 0.86 traceability.

TECHNICAL SKILLS

Languages : C++, C#, Python, Java, SQL, Blueprint (UE5)

Applied AI : local LLMs (Ollama), MCP (Model Context Protocol), structured extraction and model-output verification, agent-driven editor scripting, agent-behaviour evaluation

Engines & frameworks : Unreal Engine 5 (5.7 / 5.8), Unity 6, FastAPI, Spring Boot, .NET

Quality & tooling : regression suites, property-based testing, mutation testing, CI/CD (GitHub Actions), playtest agents, Git (GitHub, Perforce)

Transferable : team management, remote-work autonomy, technical English, classified documentation

EDUCATION & CERTIFICATIONS

Bachelor's degree in Video Game Development — [E-Artsup](#)

2020 – 2023

RNCP Level 6 certification — specialisation in C# / Unity / Game & Level Design • French Vocational Baccalaureate in Commerce — with honours

ADDITIONAL INFORMATION

Spoken languages : French (native) • English: fluent technical reading and writing, working professional speaking

Other : French nationality • interests: video games, dance, boxing, fitness.